

Muhammed Rashal M

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Summary

Full Stack Software Engineer specializing in 3D Interactive Systems and Real Time Applications, with strong frontend development and scalable backend architecture, contributing to the Research and Development department leveraging React.js, Node.js, Django, WebGL, Blender, and Linux based systems.

Experience

Full Stack Developer

January 2025 – Present

SMEC Automation Pvt Ltd

Infopark ,Ernakulam

- Built high-performance 3D web apps and digital-twin dashboards for real-time machinery visualization, smart part comparison, advanced animations, and interactive controls.
- Integrated vibration analytics and ML-based failure prediction into 3D environments, enabling early detection of mechanical issues through visual highlights and analytical graphs.
- Engineered complex backend infrastructures and efficient data pipelines that support large-scale monitoring, analytics, and visualization optimized to run even on low-resource devices like Raspberry Pi.
- I have a strong understanding of frontend design, with a focus on creating clean, user-friendly interfaces that feature effective layouts and responsive design. I am skilled in using Figma for UI/UX design and Blender for 3D asset creation, transforming complex visuals into simple, practical components.
- As part of the R&D team to experiment with multiple frameworks and technologies, delivering innovative prototypes and full-stack production solutions using Node.js, React, Prisma, Django, AWS (EC2/S3) and Linux with robust deployment, cloud integration, and high system reliability.

Education

University Of Calicut

Jun 2021 - Jun 2024

BSc Computer Science

Al Jamia College

Relevant Coursework: Data Structures, Algorithms, OS, DBMS, CN

Skills

Languages: C, C++, JavaScript, Python, HTML5, CSS3

Technologies: Microservices, REST APIs, Prisma, WebSocket, Redis, MySQL, AWS, Linux, Git, GitHub

Frameworks & Libraries: React.js, TypeScript, Tailwind CSS, Material UI, Bootstrap, Three.js, Node.js, Django, Gsap,

WebXR, GLSL Shader

Tools: Figma, Blender, Adobe softwares

Projects

- 3D Visualizer(2024): Created an interactive web-based 3D visualization tool with complex animations, real material and lighting shaders, and rendered scenes optimized for product demonstration and detailed car model visualization with the help of Three.js, WebGL, React Three Fiber, and Blender.
Live Links : [GPU-BASED FLUID FOG/CLOUD](#) | [3D CAR VISUALIZER](#)
- College Management System (2023): Web platform for course registration, study materials, program info, and attendance tracking using Django, React, JavaScript, Python, and Bootstrap.

Certification

- CS50: Introduction to Computer Science — Harvard (edX)
- Unity Essentials Pathway — Unity